

ESTHER 1, AB, Canada

WMO: 712400

Lat: 51.670N

Lon: 110.206W

Elev: 2320

StdP: 13.51

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.0	(c) -19.5	(d) -35.3	(e) 0.8	(f) -22.1	(g) -28.0	(h) 1.2	(i) -18.4	(j) 29.5	(k) 25.6	(l) 26.5	(m) 27.3	(n) 6.2	(o) 240	(p) 0.726

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 25.6	(c) 88.5	(d) 63.5	(e) 84.6	(f) 62.5	(g) 80.9	(h) 61.3	(i) 67.5	(j) 81.1	(k) 65.2	(l) 78.4	(m) 63.2	(n) 76.5	(o) 11.5	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 62.9	(b) 93.6	(c) 73.0	(d) 60.5	(e) 85.8	(f) 69.8	(g) 58.4	(h) 79.5	(i) 67.7	(j) 33.3	(k) 81.1	(l) 31.3	(m) 78.6	(n) 29.8	(o) 76.6	(p) 77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 25.9	(b) 22.9	(c) 20.4	(e) -32.0	(f) 94.6	(g) 6.0	(h) 3.5	(i) -36.3	(j) 97.0	(k) -39.8	(l) 99.1	(m) -43.2	(n) 101.0	(o) -47.5	(p) 103.5		
			DB													
			WB													
				(e) -31.9	(f) 70.7	(g) 5.8	(h) 3.2	(i) -36.1	(j) 73.0	(k) -39.5	(l) 74.9	(m) -42.8	(n) 76.7	(o) -47.0	(p) 79.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.3	9.8	12.9	24.7	39.9	51.5	59.0	65.2	63.3	53.9	40.3	25.0	13.0	
	DBStd	22.69	16.31	14.32	14.54	9.91	8.13	6.11	5.74	6.40	8.50	9.49	12.99	15.14	
	HDD50	5757	1247	1040	785	323	81	5	0	2	54	320	752	1148	
	HDD65	9892	1712	1460	1250	753	424	197	67	109	342	765	1200	1613	
	CDD50	1502	0	0	1	20	126	276	471	415	172	20	1	0	
	CDD65	161	0	0	0	0	4	17	73	57	10	0	0	0	
	CDH74	2956	0	0	1	9	155	315	1143	1005	318	10	0	0	
	CDH80	971	0	0	0	0	29	73	405	362	102	0	0	0	
Wind	WSAvg	9.7	9.2	9.3	10.3	10.9	10.9	9.8	9.4	8.8	9.4	10.0	9.5	9.5	
Precipitation	PrecAvg	13.0	0.6	0.3	0.5	0.8	1.5	2.9	2.1	1.7	1.0	0.6	0.5	0.4	
	PrecMax	18.5	1.9	0.7	1.4	2.4	3.1	5.9	5.0	3.3	3.3	1.3	1.7	0.7	
	PrecMin	6.8	0.1	0.0	0.1	0.1	0.1	0.7	0.8	0.2	0.0	0.2	0.0	0.1	
	PrecStd	3.0	0.4	0.2	0.3	0.6	0.9	1.3	0.9	0.8	0.8	0.3	0.4	0.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.8	44.4	60.5	74.8	83.6	86.9	93.7	93.1	89.3	74.6	58.7	47.0	
		MCWB	38.0	38.1	45.7	52.7	57.4	63.6	67.8	63.9	60.4	54.3	45.6	38.7	
	2%	DB	40.0	39.3	53.2	68.5	78.4	81.5	88.7	88.2	82.3	66.6	51.1	41.9	
		MCWB	35.2	34.7	41.9	50.1	55.8	61.0	65.6	62.4	58.3	50.6	42.1	36.2	
	5%	DB	36.9	36.2	47.6	62.6	73.5	77.5	84.7	84.0	76.8	60.9	45.6	38.2	
		MCWB	33.1	32.3	39.4	46.4	54.2	59.9	64.2	62.0	56.9	48.1	38.8	33.9	
	10%	DB	32.9	32.6	42.6	57.6	68.8	73.5	80.4	79.8	71.4	55.7	41.0	33.8	
		MCWB	29.9	29.6	36.7	44.2	51.8	58.2	63.0	61.1	54.5	45.1	36.0	30.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.5	38.3	47.3	54.4	61.1	67.8	72.7	68.7	62.6	55.8	46.7	40.4
			MCDB	43.4	43.7	58.2	71.3	74.7	81.0	86.4	83.9	82.0	72.8	57.5	45.4
2%		WB	35.6	35.1	42.8	50.8	58.0	64.2	69.3	65.9	59.9	51.6	42.9	36.6	
		MCDB	39.4	38.8	51.4	65.7	73.4	75.8	83.0	80.6	78.3	64.1	50.2	41.0	
5%		WB	33.2	32.5	39.8	47.8	55.6	61.6	66.4	63.9	57.7	48.5	39.4	33.9	
		MCDB	36.7	35.7	46.6	60.5	70.7	73.1	78.9	78.2	74.3	59.3	45.1	38.4	
10%		WB	29.9	29.6	37.1	45.0	53.2	59.6	64.3	62.0	55.4	45.9	36.5	30.6	
		MCDB	32.8	32.5	42.5	55.6	66.6	70.6	77.2	76.3	69.9	54.6	40.7	33.9	
Mean Daily Temperature Range		5% DB	MDBR	18.0	18.8	19.6	23.5	24.9	22.5	25.6	26.3	25.6	22.2	17.6	17.6
			MCDBR	19.0	20.6	26.0	31.9	32.4	29.2	32.8	34.9	36.2	31.2	22.9	20.1
	MCWBR		15.5	16.8	16.1	16.4	15.1	13.1	13.3	13.9	16.2	17.3	15.3	15.5	
	5% WB	MCDBR	18.9	20.2	24.2	29.5	29.0	25.8	29.0	29.8	33.4	29.5	22.1	20.1	
		MCWBR	15.9	16.8	16.1	16.1	14.3	13.0	13.5	13.4	15.7	16.9	15.3	15.9	
Clear-Sky Solar Irradiance	taub	0.259	0.295	0.300	0.351	0.379	0.373	0.371	0.386	0.331	0.291	0.263	0.251		
	taud	2.340	2.279	2.414	2.294	2.255	2.333	2.370	2.293	2.427	2.515	2.463	2.353		
	Ebn at Noon	244	262	287	281	277	279	278	266	272	266	244	228		
	Edh at Noon	19	27	29	37	41	38	36	37	29	22	17	16		
All-Sky Solar Radiation	RadAvg	367	704	1141	1532	1854	1917	2055	1669	1198	722	378	273		
	RadStd	36	60	88	129	127	128	91	128	111	93	35	27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page